MILITARY SPECIFICATION SHEET

ELECTRON TUBE, GAS

TYPE 3B22

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Rectifier, full-wave

See figure 1

Mounting position: Any

Weight: 2.5-ounces (70.88 grams) nominal

ABSOLUTE RATINGS:

Parameter:	Ef	epx	Epp	ib	*isurge* 0.1 sec	Io	F	tk	TA
Unit:	Vac	v	Vac	a	a	Adc	Hz	sec	°C
Maximum:	2.625	725	---	4.0	60	1.0	150	---	+70
Minimum:	2.375	---	---	---	---	---	---	20	-55
TEST CONDITIONS:	2.5	---	270	---	---	---	60	20	---

GENERAL:

Qualification - Required

Ⓓ denotes changes

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS		UNIT
							MIN	MAX	
	Quality conformance inspection, part 1								
1231	Peak emission by voltage drop	1	ib = 4.0 a	0.65	II	etd	---	18.0	v
1256	Electrode current (anode)	1, 2	Ebb = 20 Vdc; Rp = 20 ohms	0.65	II	Ib	100	---	mAdc
1301	Filament current	-		0.65	II	If	5.5	7.0	Aac
1353	Operation of rectifiers	3	ib = 4.0 a (min); Io = 1.0 Adc; t = 120 minutes	0.65	II	---	---	---	---
	Quality conformance inspection, part 2								
(D) 1042	Shock, specified pulse	-		---	---	---	---	---	---
	Quality conformance inspection, part 3								
---	Life test	-	Group C; operation of rectifiers; t = 500 hours	---	---	---	---	---	---
---	Life-test end point:								
1231	Peak emission	-		---	---	etd	---	20.0	v

NOTES:

1. Test each unit separately.
2. Record emission readings 180-seconds minimum after test voltage has been applied.
3. This test shall be the first test performed at the conclusion of the holding period.

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Air Force - 11, 17, 80
DSA - ES

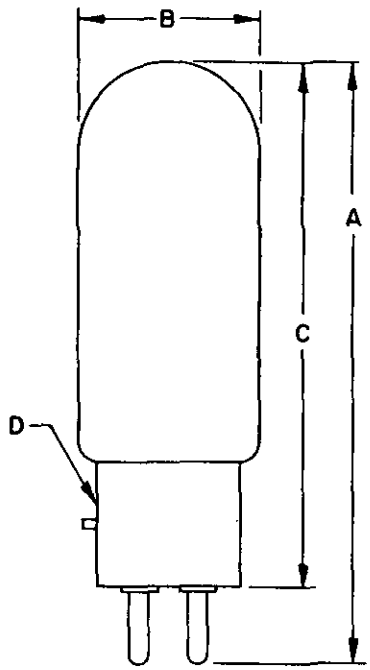
User activities:
Army - MU
Navy - AS, OS, MC, CG, SH
Air Force - 19

Preparing activity:
Navy - EC

Agent:
DSA - ES

(Project 5960-2933)

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PIN CONNECTIONS	
Pin No.	Element
1	f
2	2a
3	1a
4	f

Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Quality conformance inspection, part 2		
A	5.000 (127.00)	6.000 (152.40)
B	1.438 (36.53)	1.703 (43.26)
C	4.375 (111.13)	
Ⓓ Quality conformance inspection, part 3 (see note)		
D	Base: A4-10 (EIA)	

NOTE:
This dimension shall be tested annually with the following sampling plan:

n1 = 4	c1 = 0	where c2 represents the total allowable failures
n2 = 4	c2 = 1;	for the first and second samples combined.

In case of failure after double sampling, the failing dimension(s) shall become quality conformance inspection (QCI), part 2, for three successful consecutive submissions, at which time the test may revert to the QCI, part 3, basis.

FIGURE 1. Outline drawing of electron tube type 3B22.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

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   ☐ YES ☐ NO *(If "Yes", in what way?)*

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DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED S/N 0102-014-1802

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